# Functional Requirement Document (FRD)

**Project:** E-commerce Platform for Electronic Gadgets

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## 1. Introduction

This Functional Requirement Document (FRD) provides a detailed specification of the features and functionality for the e-commerce platform outlined in the BRD. It expands on the high-level business requirements by defining use cases, user stories, and acceptance criteria that will guide design, development, and testing activities.

## 2. Functional Areas

### 2.1 User Registration & Authentication

**Use Case:** New User Registration

- **Actors:** Visitor, System
- **Preconditions:** User is not registered and has access to a valid email or social account.
- **Flow:**
    1. User clicks on "Sign Up" option.
    2. System displays registration form (name, email, password, confirm password).
    3. User submits form; system validates inputs.
    4. System sends a verification link to the email.
    5. User verifies and gains access to account.

**User Story:**
*As a new user, I want to create an account so that I can access personalized features and make purchases.*

**Acceptance Criteria:**

- Registration form must validate unique email.
- Verification email should be sent upon registration.
- User should be able to log in after successful verification.

### 2.2 Product Browsing & Search

**Use Case:** Product Search and Filter

- **Actors:** Registered/Guest User, System
- **Preconditions:** Product catalog exists.
- **Flow:**
    1. User enters keyword in search bar.
    2. System retrieves relevant results.
    3. User applies filters (price, brand, category).
    4. Results update dynamically.

**User Story:**
*As a customer, I want to search and filter products so that I can quickly find items that match my needs.*

**Acceptance Criteria:**

- Search must return relevant results within 2 seconds.
- Filters should work in combination (e.g., price + brand).
- Recently viewed/recommended products should be displayed.

### 2.3 Product Details

**Use Case:** View Product Information

- **Actors:** Registered/Guest User
- **Flow:**
    1. User clicks on a product.
    2. System displays details (images, specifications, price, stock availability).
    3. User can view customer ratings and reviews.
    4. Option to view similar or related products.

**User Story:**
*As a shopper, I want to see detailed product information so that I can make an informed purchase decision.*

**Acceptance Criteria:**

- Product pages must include at least 3 images.
- Pricing should clearly show discounts (if applicable).
- Customers can add or read reviews.

### 2.4 Shopping Cart & Checkout

**Use Case:** Checkout Process

- **Actors:** Registered User, System, Payment Gateway
- **Flow:**
    1. User adds product to cart.
    2. User proceeds to checkout.
    3. System displays order summary and shipping address.
    4. User selects payment method and confirms.
    5. Payment gateway processes transaction.
    6. System displays order confirmation.

**User Story:**
*As a customer, I want a secure and simple checkout process so that I can place orders with confidence.*

**Acceptance Criteria:**

- Cart must allow adding/removing items.
- Multiple payment options (credit, debit, UPI, wallets) must be available.
- Confirmation email/SMS must be sent after order.

### 2.5 Order Management

**Use Case:** Track Order Status

- **Actors:** Registered User, System
- **Flow:**
    1. User navigates to “My Orders.”
    2. System displays order history.
    3. User selects an order to see tracking details.
    4. System updates status (Processing, Shipped, Delivered).

**User Story:**
*As a customer, I want to track my order so that I know when it will arrive.*

**Acceptance Criteria:**

- Order status must update in real-time.
- Tracking link should be available once dispatched.
- Users should be able to cancel orders before shipping.

### 2.6 User Account Management

**Use Case:** Manage Profile & Addresses

- **Actors:** Registered User, System
- **Flow:**
    1. User logs in to account.
    2. User updates personal details or adds shipping addresses.
    3. System validates and saves changes.

**User Story:**
*As a registered user, I want to manage my profile and addresses so that I can shop more efficiently.*

**Acceptance Criteria:**

- Users can update profile info anytime.
- Multiple shipping addresses should be allowed.
- Order history should be accessible.

## 3. Non-Functional Requirements

- The platform should handle at least 500 concurrent users.
- Response time for key operations (search, add to cart) should be < 2 seconds.
- System must comply with PCI DSS for payments and SSL for data security.
- The application should work seamlessly across modern browsers and mobile devices.